

HONGHAO CAO

MIT EECS PhD Candidate | Optical & Imaging Systems

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ABOUT ME

Optics researcher and imaging systems engineer specializing in end-to-end precision instruments that integrate optical design, high-speed electronics, synchronized control, and machine-learning-based image analysis. Experienced in building complete systems from hardware architecture through data acquisition and reconstruction. Particularly interested in translating this systems expertise to robust visual and optical perception for camera and display systems, robotics, and AR/VR.

RESEARCH PROJECTS

High-Speed Three-Photon Fluorescence Lifetime Microscope

2025 – Present

End-to-end optical instrument for high-speed, sensitive, and specific multiphoton imaging

- Built a three-photon fluorescence lifetime microscope integrating femtosecond excitation, galvo scanning, hybrid photodetection, high-speed digitization, and synchronized motion and control electronics.
- Developed a high-speed waveform acquisition and processing architecture operating at 5 GS/s and 7 GB/s, enabling fluorescence lifetime imaging in a single shot.

High-Throughput Imaging System

2024 – 2026

A robust nonlinear pencil beam for volumetric imaging

- Developed a self-localized ultrafast pencil beam that extends the depth of focus by 10× while preserving near-diffraction-limited lateral resolution, enabling substantially higher-throughput volumetric imaging.
- Integrated the beam into a high-NA (NA=1.05) multiphoton microscope, reducing acquisition time from 25 min to 1 min for a 2 mm × 2 mm × 50 μm volume while maintaining subcellular resolution.

Adaptive Ultrafast Optical System

2022 – 2024

Programmable optical hardware for spectral, temporal, and spatial light-field control

- Developed an integrated optical platform combining femtosecond lasers, programmable optomechanics, spectroscopy, autocorrelation, beam imaging, and automated control for real-time optimization of broadband optical output.
- Demonstrated wide-range control of optical spectrum, pulse duration, and beam profile across two optical octaves, reaching 0.72 MW peak power and enabling up to 15× higher nonlinear imaging signal.

SELECTED PUBLICATIONS

Self-localized ultrafast pencil beam for volumetric multiphoton imaging

Featured on the May 2026 journal cover

Nature Methods 23, 1024-1036 (2026). Cao, H., Spitz, S., Yu, L., Liu, K., Zhang, Z., Presutti, F., ..., You, S.

Spectral-temporal-spatial customization via modulating multimodal nonlinear pulse propagation

Nature Communications 15(1), 2031 (2024). Qiu, T*, Cao, H*, Liu, K., Yu, L., ..., You, S. (*equal contribution)

EDUCATION

Massachusetts Institute of Technology

09/2024 – Present | Cambridge

Ph.D., Electrical Engineering and Computer Science

Massachusetts Institute of Technology

09/2022 – 06/2024 | Cambridge

S.M., Electrical Engineering and Computer Science

GPA: 5.0/5.0

Huazhong University of Science and Technology

09/2018 – 06/2022 | Wuhan

B.E., Optical and Electronic Information

GPA: 3.99/4.0, Rank: 1/245